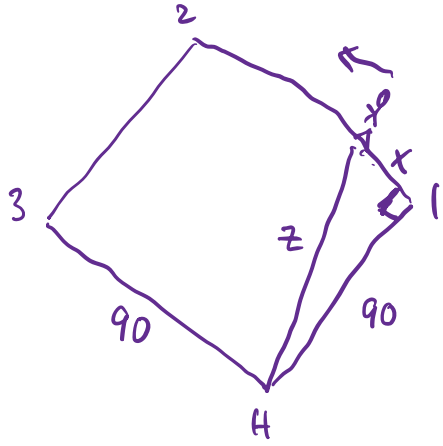
Problem 10. TIER 2 — NAMING THE VARIABLES IS THE PROBLEM

A baseball diamond is a 90 ft square. A runner heads from first base toward second at 24 ft/s. How fast is his distance **from home plate** changing when he is 30 ft from first?



(1)

(2) $\frac{dz}{dt} = ?$, $\frac{dx}{dt} = 24$ when $x = 30$

(3) $x^2 + 90^2 = z^2$

(4) $\frac{d}{dt}(x^2 + 90^2) = \frac{d}{dt} z^2$

$2x \frac{dx}{dt} = 2z \frac{dz}{dt}$

$\frac{dz}{dt} = \frac{x}{z} \frac{dx}{dt}$

(5) $x^2 + 90^2 = z^2$

$30^2 + 90^2 = z^2$

$z^2 = 900 + 8100 = 9000$

$z = \sqrt{9000} = \sqrt{10 \cdot 900} = \sqrt{10}\sqrt{900} = 30\sqrt{10}$

$\frac{dz}{dt} = \frac{30}{30\sqrt{10}} 24 = \frac{24}{\sqrt{10}} \frac{\sqrt{10}}{\sqrt{10}} = \frac{24\sqrt{10}}{10} = \frac{12\sqrt{10}}{5} \frac{ft}{sec}$

$\approx 7.59 \text{ ft/sec}$